

Yoga Pose Detection and Correction System

A PROJECT REPORT

For

“CSE 456: DIGITAL IMAGE PROCESSING”

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SECTION:M

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in

COMPUTER SCIENCE AND ENGINEERING

of

SCHOOL OF ENGINEERING AND SCIENCES



SRM University-AP, Neerukonda, Andhra Pradesh 522240

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DECLARATION

We, the undersigned students of B.Tech.– Computer Science & Engineering, hereby declare that the **CSE456: DIGITAL IMAGE PROCESSING Project** titled :
“Yoga Pose Detection and Correction System”

submitted in partial fulfillment of the requirements for the End Semester Examination Evaluation, is an authentic work carried out by us under the guidance of **Dr. Bojja Venu**

We further declare that:

1. This project is the result of our own work and has not been submitted to any other university or institute for any academic purpose.
2. All information, data, diagrams, and results presented in this project are true to the best of our knowledge and belief.
3. Proper acknowledgments have been given wherever external sources have been used.
4. We understand that any violation of academic integrity may lead to disciplinary action as per institute norms.

Date:

Place:

Sl. No.	Name of Student	Registration No.	Signature
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FACULTY GUIDE'S CERTIFICATION

I hereby certify that the above declaration made by the students is true, and the project titled

“Yoga Pose Detection and Correction System “ has been completed under my supervision.

Signature of Supervisor: _____

Name: Dr. Bojja Venu

Designation: Assistant Professor

**Department: CSE, SEAS
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**University-
AP**

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INTRODUCTION

Yoga is widely practiced to improve physical health, flexibility, and mental well-being. However, performing yoga poses incorrectly can reduce effectiveness and may lead to injuries. Maintaining proper posture requires proper alignment of body joints, which is difficult to achieve without expert guidance.

This project focuses on developing a Yoga Pose Detection and Correction System that uses computer vision and pose estimation to analyze human posture in real time. By detecting body landmarks and calculating joint angles, the system evaluates pose accuracy, provides corrective feedback, and helps users improve posture during yoga practice.

PROBLEM STATEMENT:

Many people practice yoga without professional supervision, leading to incorrect postures and possible injuries. Traditional self-practice lacks instant feedback and correction.

This system aims to solve this problem by:

- Continuously analyzing user posture
- Detecting incorrect joint alignment
- Providing real-time feedback and correction suggestions

OBJECTIVES:

The main objectives of this system are:

- To detect yoga poses in real time
- To extract body landmarks and compute joint angles
- To compare user posture with a reference pose
- To calculate pose accuracy score
- To provide corrective feedback for posture improvement

PROPOSED SOLUTION

2.1 Core Idea

The proposed solution is a computer vision based yoga pose evaluation system that analyzes user posture using pose estimation and compares it with an ideal reference pose.

The system extracts meaningful features such as:

- Body landmarks
- Joint angles
- Pose similarity scores
- Posture deviation measurements

These features are used to determine posture correctness.

2.2 Working of the System

1. Input Collection

The system takes:

- Reference yoga pose image
- Live webcam input from user

2. Pose Detection

Using MediaPipe Pose:

- Detects body landmarks
- Tracks major joints
- Extracts pose coordinates

3. Angle Calculation

Calculates joint angles for:

- Elbows
- Shoulders
- Hips
- Knees

Using geometric angle computation.

4. Pose Comparison

Compares user pose with target pose using:

- Angle difference
- Cosine similarity

Generates similarity score.

5. Feedback and Output

Classifies posture as:

- Correct
- Slightly Incorrect
- Incorrect

Provides corrective suggestions.

2.3 Correction Strategy

The system offers preventive guidance by:

- Highlighting incorrect joints
- Suggesting arm/leg adjustments
- Providing real-time posture correction

2.4 Key Innovation

The uniqueness of this system lies in:

- Real-time yoga guidance
- Angle-based posture evaluation
- Explainable feedback system
- Lightweight training-free approach

2.5 Expected Outcome

The system helps:

- Improve yoga posture accuracy
- Assist beginners without trainers
- Reduce posture-related injuries
- Promote effective self-practice

IMPLEMENTAION & RESULTS

IMPLEMENTAION DETAILS:

3.1 Technology Stack

Programming Language: Python

Pose Estimation: MediaPipe

Computer Vision: OpenCV

Numerical Processing: NumPy

Similarity Computation: SciPy

3.2 Pose Detection Implementation

The system uses body landmarks generated by MediaPipe.

Tracks:

- Shoulders
- Elbows
- Hips
- Knees

for pose evaluation.

3.3 Angle Calculation Implementation

Joint angle formula:

$$\theta = \text{atan}((y_c - y_b)/(x_c - x_b)) - \text{atan}((y_a - y_b)/(x_a - x_b))$$

Used for posture analysis.

3.4 Feature Extraction

The system extracts:

- Joint angles
- Landmark coordinates
- Similarity scores
- Pose deviation values

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3.5 Pose Comparison Model

Compares:

- User pose vector
- Reference pose vector

Using:

$$\text{Similarity} = (A \cdot B) / (|A||B|)$$

3.6 Score Calculation

Pose accuracy score computed from:

- Angle error
- Landmark similarity

Classification:

0–50% → Poor

50–75% → Moderate

75–90% → Good

90–100% → Excellent

3.7 Output Generation

The system generates:

- Pose label
- Accuracy score
- Incorrect joint highlights
- Correction suggestions

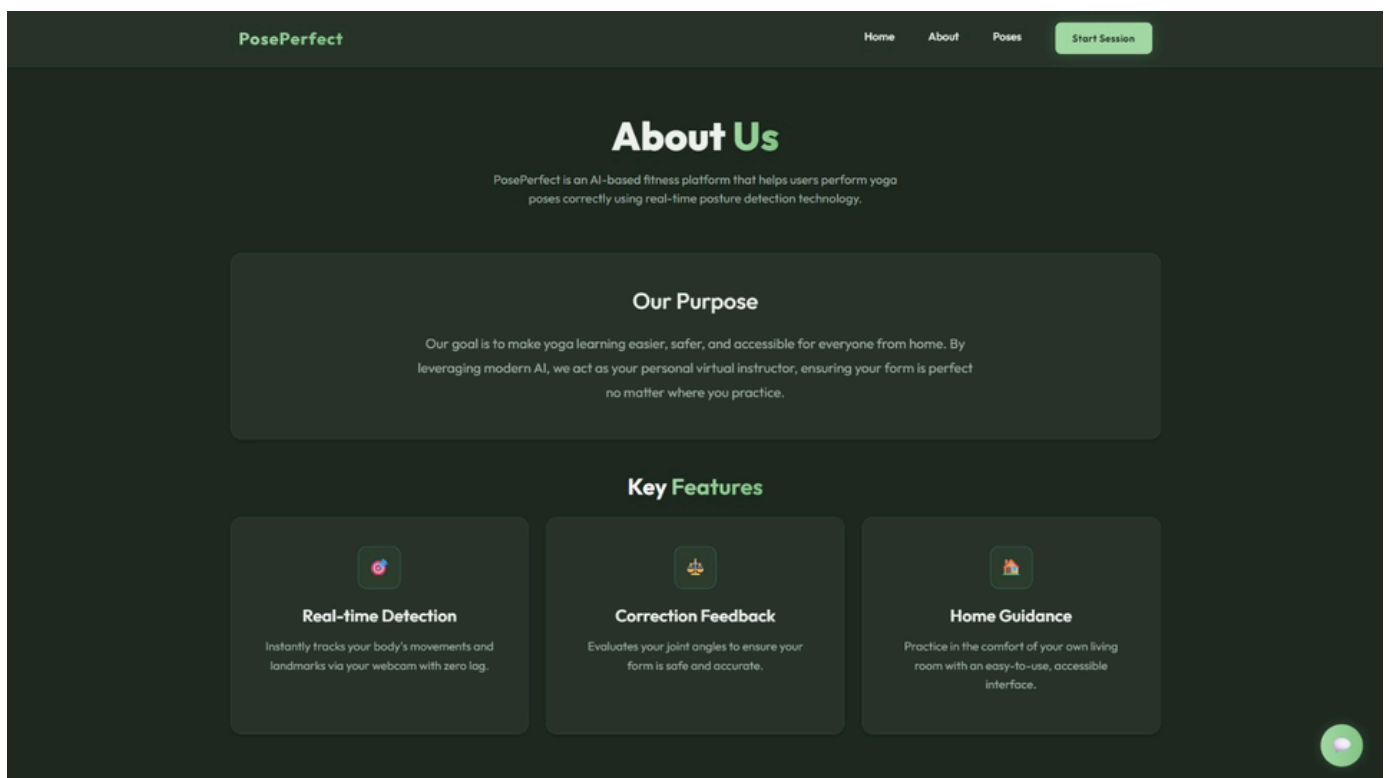
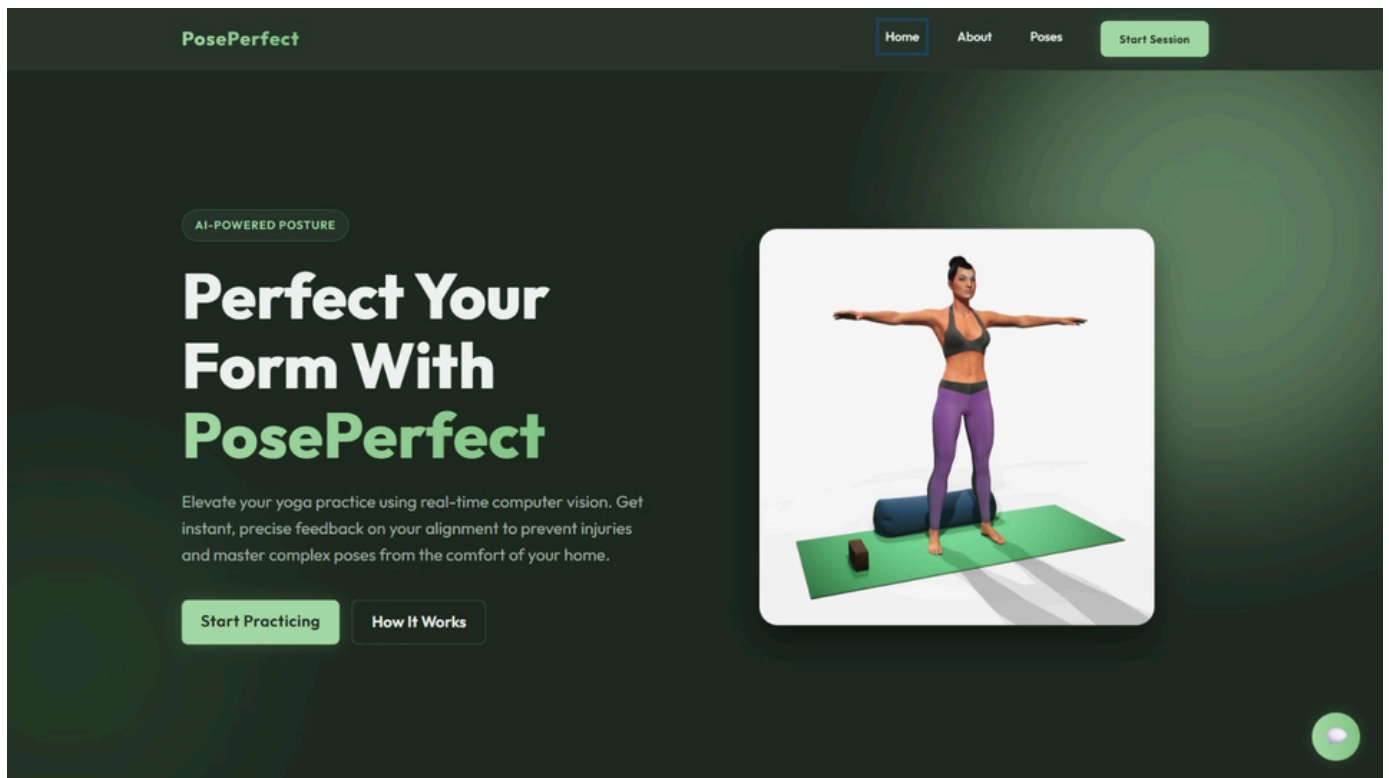
3.8 System Workflow

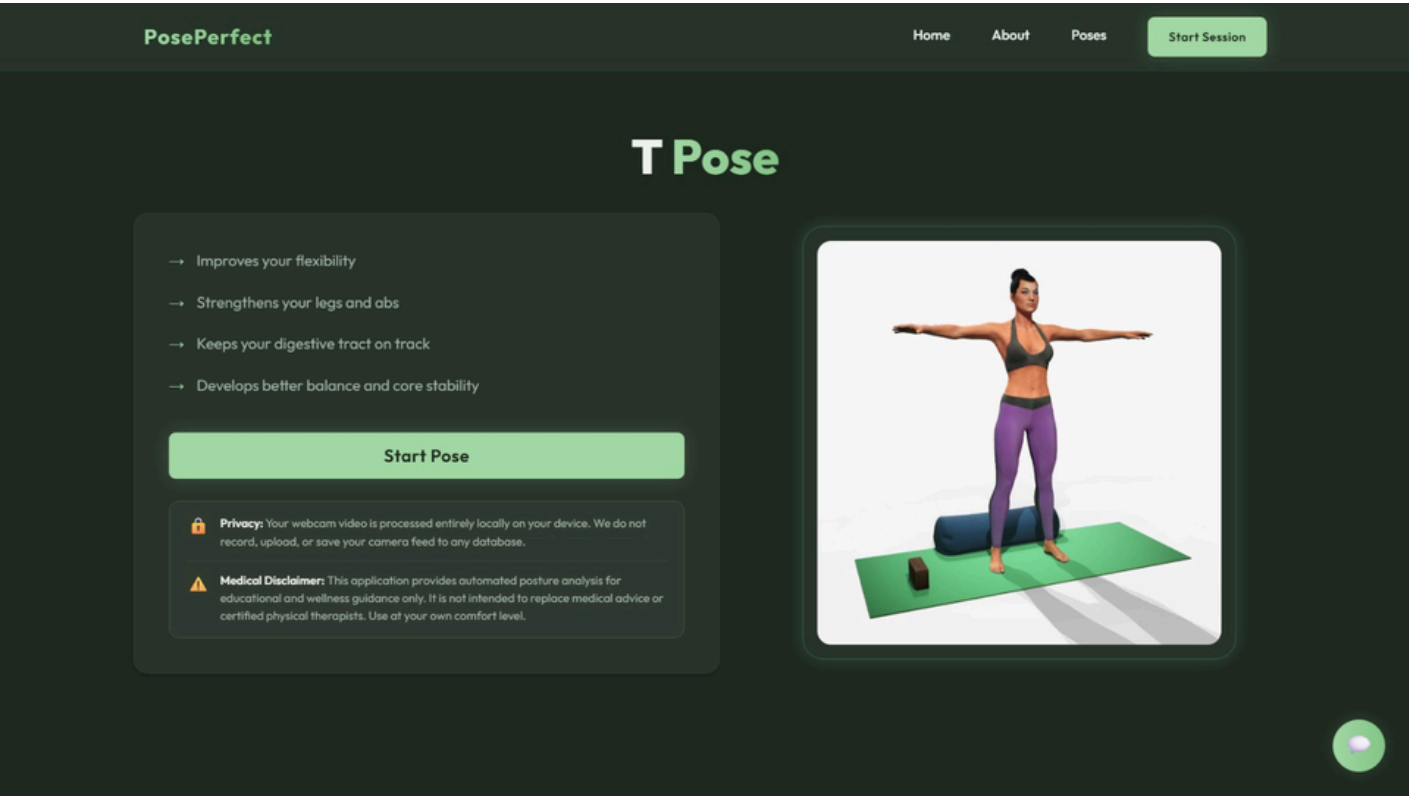
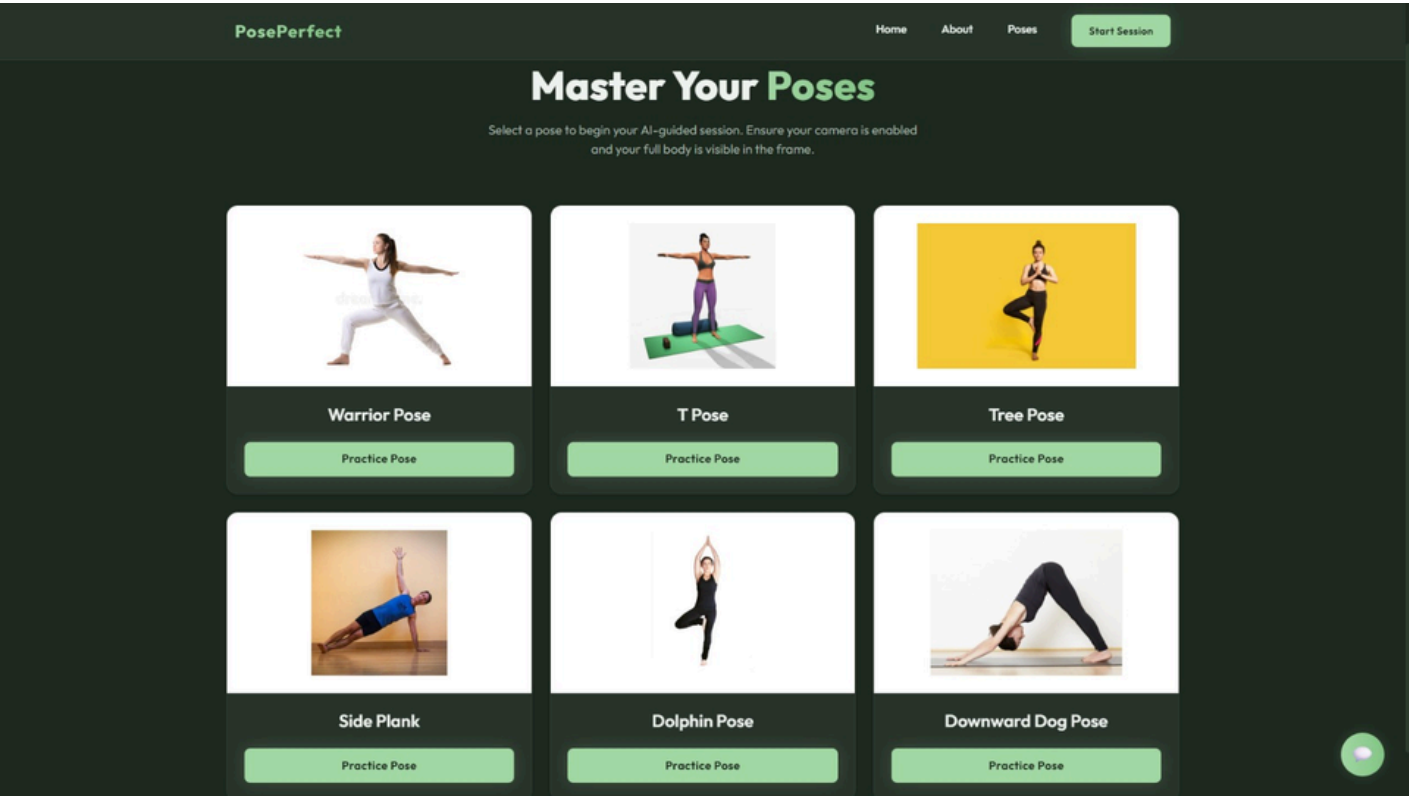
1. Capture user input
2. Detect pose
3. Extract landmarks
4. Calculate angles
5. Compare with target pose
6. Generate score
7. Display feedback

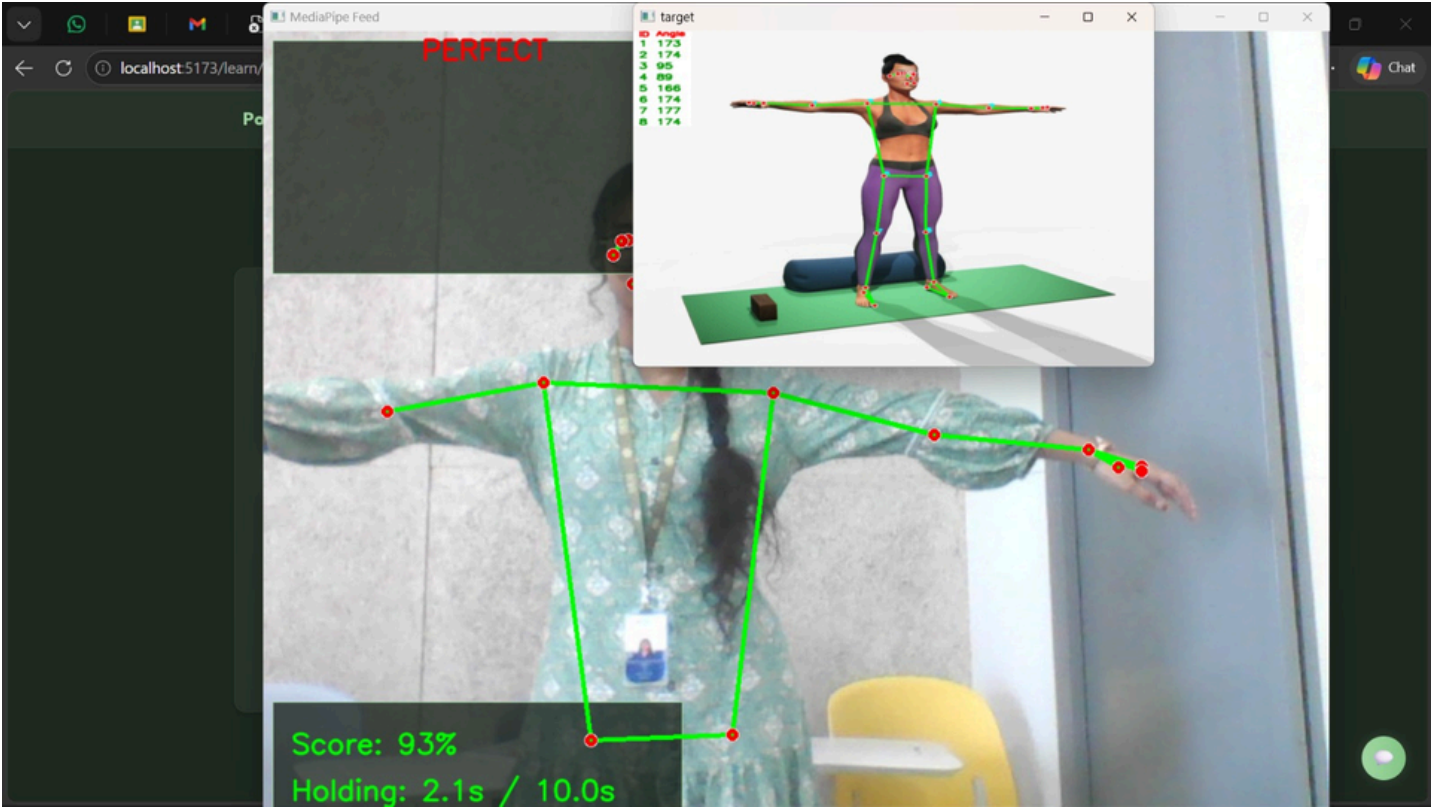
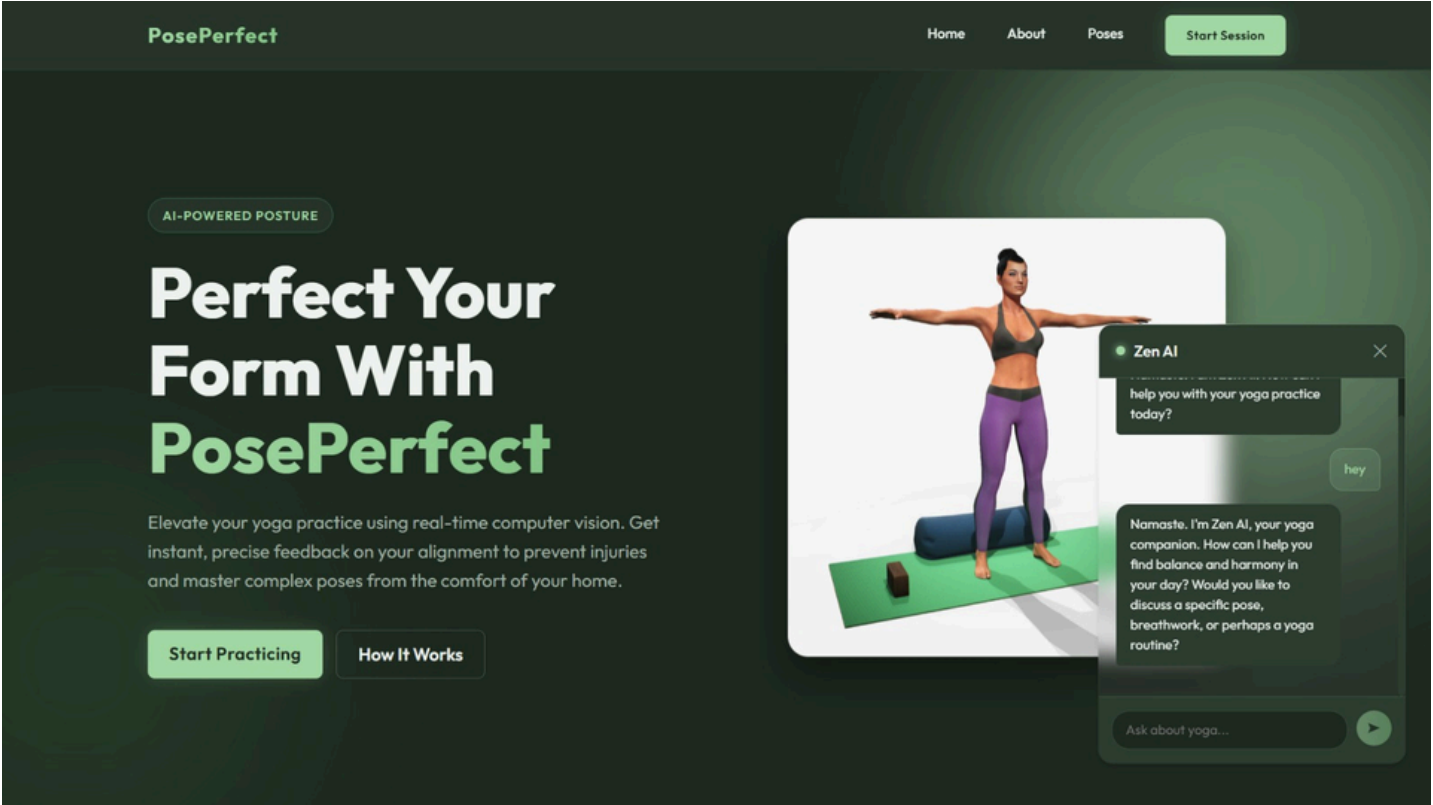
3.9 Performance Considerations

- Lightweight implementation
- Real-time processing
- Low computational cost
- Scalable for more poses

RESULTS:







OBSERVATIONS:

- The system successfully detects and analyzes yoga poses in real time.
- Joint angle calculations effectively identify deviations from the reference pose.
- The angle-based comparison method provides reliable posture evaluation.
- Real-time feedback helps users correct arm, leg, and body alignment.
- Cosine similarity and angle error together improve pose accuracy assessment.
- The system performs well for predefined yoga poses under proper conditions.
- Accuracy depends on factors such as lighting, camera angle, and user positioning.
- Complex poses or partial body occlusions may reduce detection precision.
- Overall, the system demonstrates practical and effective guided yoga assistance.

FUTURE WORKS

- Improve accuracy using advanced deep learning models
- Support a wider range of yoga poses with a larger dataset
- Implement 3D pose estimation for better joint analysis
- Develop a mobile application for easy accessibility
- Add real-time voice guidance for user assistance
- Enhance performance under varying lighting and backgrounds
- Introduce personalized feedback based on user ability
- Optimize system for faster real-time processing

CONCLUSION

The Yoga Pose Detection and Correction System presents an effective application of computer vision and pose estimation for improving yoga practice through automated guidance. By using MediaPipe Pose to detect body landmarks and calculating joint angles for posture evaluation, the system can analyze user poses in real time and compare them with a reference posture. This enables accurate scoring and meaningful corrective feedback, helping users perform yoga poses more accurately and safely.

A major strength of the system is its ability to provide both pose evaluation and real-time correction suggestions, making it useful for beginners practicing without an instructor. The use of geometric analysis and similarity measures makes the system lightweight, interpretable, and efficient without requiring complex training models. The project demonstrates how computer vision can be applied in health and fitness assistance at low cost.

Although the system performs well for supported poses, its accuracy may be affected by environmental factors and camera orientation. In future, the system can be enhanced by supporting more poses, improving robustness, and integrating personalized or AI-based guidance. Overall, the project successfully meets its objective of providing an intelligent and practical yoga posture correction assistant.

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